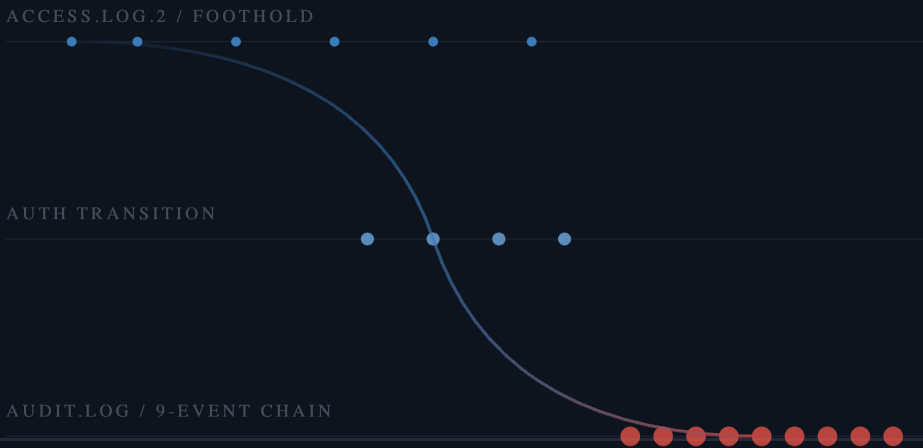


CASE FILE // AIT-LDS V2.0 / RUSSELLMITCHELL

Reconstructing a privilege escalation attack from raw logs.

A 3NF PostgreSQL schema turns an access.log.2 foothold into a 9-event su + sudo chain on intranet_server.



TEAM	INSTRUCTOR	DELIVERABLES
Naman Sudan River Roseveare-Hunt Ishaan Shetty	Dr. Liu	22-table schema 23 SQL queries 9-panel dashboard 10-section report
	STACK	
	PostgreSQL 16 · Alembic SQLAlchemy · Streamlit · Plotly	

How we built it.

GitHub



- Version control with reviewed pull requests.
- Single source of truth for schema, queries, dashboard.

`github.com/NamanSudan/data-201-security-log-analysis`

Linear



- Sprint planning against rubric line items.
- Task tracking and ownership.

Slack



- Day-to-day team communications.

Docker



- Reproducible PostgreSQL 16 environment.
- One-command database rebuild from scratch.

Alembic



- Schema migrations versioned in Git.
- Every table change ships as a migration.

Streamlit + Plotly

- Interactive dashboard layer.
- Hits PostgreSQL directly via SQLAlchemy.

STACK · POSTGRESQL 16 + SQLALCHEMY + PYTHON LOADERS FOR THE DATA PLANE.

A 22-table 3NF schema over the russellmitchell testbed.

22

HOSTS IN INVENTORY

8

SOURCE LOG FILES

61,862

LABELED ATTACK LINES

~439K

LOADED ROWS

SOURCE DATA, THREE DOMAINS

Host inventory: 7 tables, including host, fqdn, ip_address, distribution.

Audit events: 10 tables, with audit_event as the supertype and 8 subtype tables for the audit-type families.

Attack labels: 5 tables, including attack_phase, attack_label, labeled_line, plus a per-rule annotation table.

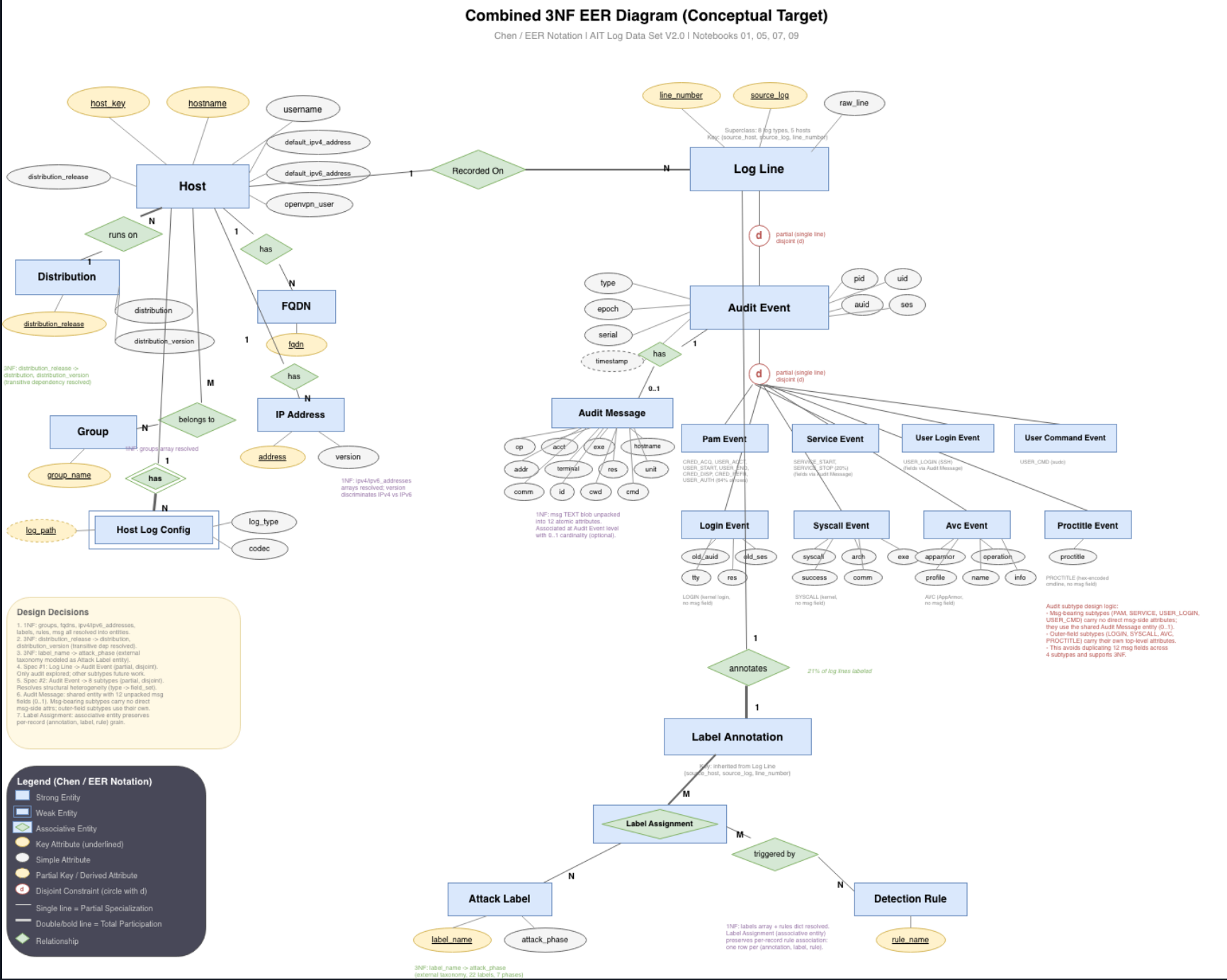
STORY SETUP

An access.log.2 foothold leads into the audit.log su then sudo chain on intranet_server.

TWO LOG FAMILIES. ONE HOST. NINE AUDIT EVENTS.

A 22-table 3NF schema over the russellmitchell testbed.

CONCEPTUAL 3NF SCHEMA · EER DIAGRAM



SCHEMA SUMMARY

22
TABLES

3
DOMAINS

3NF
NORMAL FORM

DOMAINS

Host inventory · 7 tables host, fqdn, ip_address, distribution, ...

Audit events · 10 tables audit_event supertype + 8 subtypes, audit_message

Attack labels · 5 tables attack_phase, attack_label, labeled_line, labeled_line_label, labeled_line_rule

CHEN NOTATION · PKS UNDERLINED · FK CONSTRAINTS ARROWED

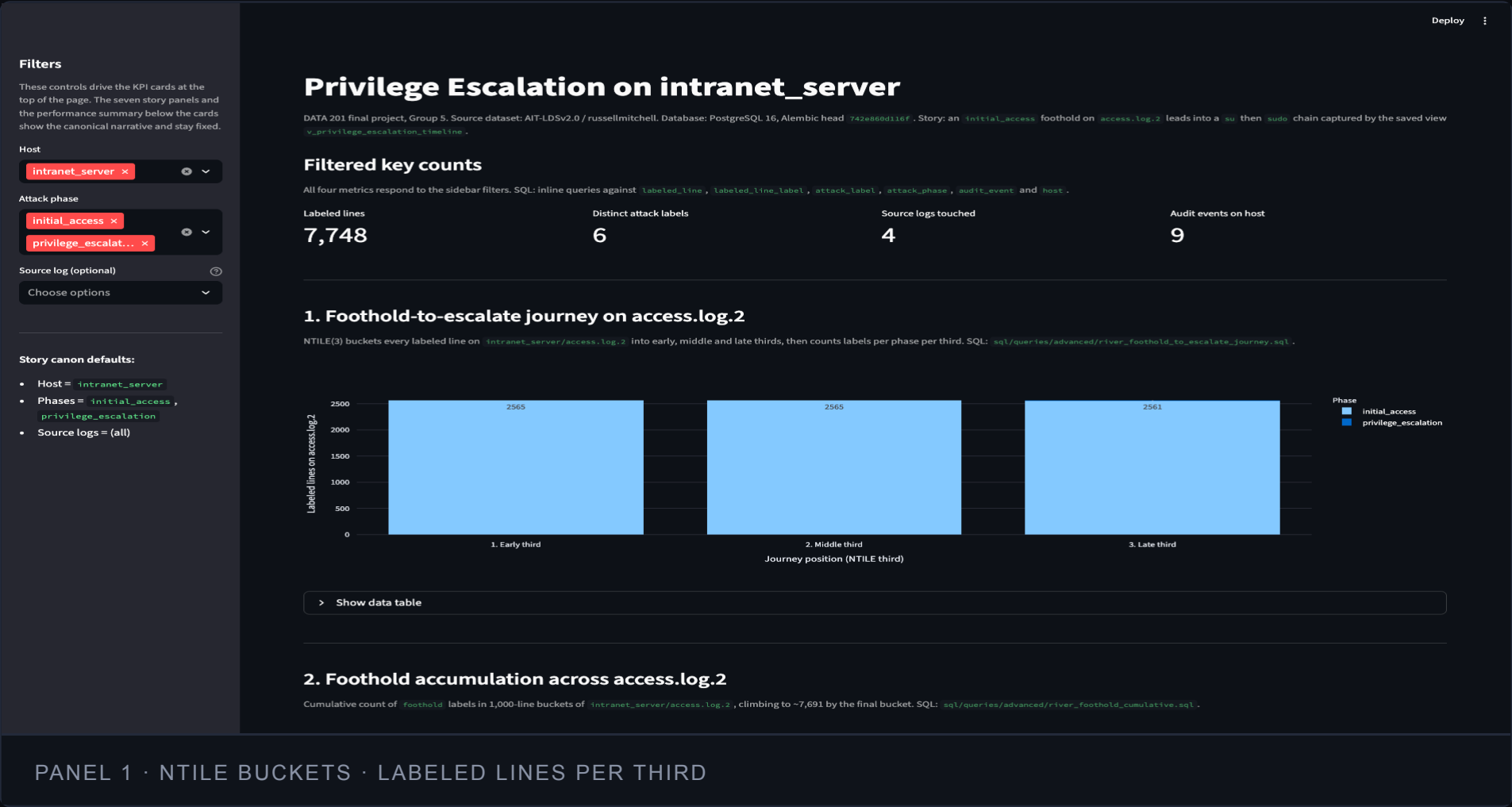
23 queries in the repo. The slides spotlight four.

Full portfolio lives in GitHub and report section 6. The spotlight rows below are the deep dives that drive the next four slides.

AUTHOR	QUERY	TECHNIQUE TAGS
River	Foothold-to-escalation journey river_foothold_to_escalate_journey.sql	CTE NTILE WINDOW
	Foothold accumulation across access.log.2	CUMULATIVE SUM WINDOW
	Source spread per host, host and label baselines	RANK JOIN ≥3 BASIC
Ishaan	Audit types in privilege escalation ishaan_audit_types_in_privilege_escalation.sql	EXISTS SUBQUERY
	Detection rules ranked, top rule per label	RANK WINDOW CORRELATED SUBQUERY
	Phase and rule baseline counts	BASIC
Naman	EXPLAIN ANALYZE + composite index naman_explain_index_improvement.sql	EXPLAIN ANALYZE INDEX
	Saved view v_privilege_escalation_timeline naman_view_privilege_escalation_timeline.sql	SAVED VIEW JOIN ≥8
	Audit-type density per phase, service baselines	DENSE_RANK WINDOW BASIC

Where in access.log.2 does the foothold turn into escalation evidence?

```
WITH access_labels AS (  
  
  SELECT ll.line_number,  
         al.label_name,  
         ap.phase_name,  
  
  NTILE(3) OVER (ORDER BY ll.line_number) AS journey_third  
  
  FROM labeled_line ll  
  
  JOIN labeled_line_label lll ON lll.labeled_line_id = ll.labeled_line_id  
  
  JOIN attack_label al ON al.label_id = lll.label_id  
  
  JOIN attack_phase ap ON ap.phase_id = al.phase_id  
  
  WHERE ll.source_host = 'intranet_server' AND ll.source_log = 'access.log.2' AND ap.phase_name IN  
    ('initial_access', 'privilege_escalation' )  
)  
SELECT journey_third, phase_name, COUNT(*) AS labeled_lines FROM access_labels GROUP BY  
journey_third, phase_name ORDER BY journey_third, phase_name;
```



Foothold spreads across all three thirds of access.log.2. Privilege-escalation evidence appears only in the late third. NTILE turns raw line position into a readable attack journey.

An 8-table saved view that becomes the privilege-escalation timeline.

```
CREATE OR REPLACE VIEW v_privilege_escalation_timeline AS SELECT h.host_key AS host,
    ae.timestamp,
    ae.type
AS audit_type,
    am.op
AS pam_operation,
    am.acct
AS target_account,
    am.exe
AS executable,
    am.terminal,

string_agg(DISTINCT al.label_name, ',' ORDER BY al.label_name) AS labels,

string_agg(DISTINCT llr.rule_name, ',' ORDER BY llr.rule_name) AS rules FROM audit_event
ae JOIN host h ON h.host_id = ae.host_id LEFT JOIN audit_message am

ON am.event_id = ae.event_id JOIN labeled_line ll

ON ll.source_host = h.host_key

AND ll.source_log = 'audit.log' AND ll.line_number = ae.line_number JOIN labeled_line_label III

ON III.labeled_line_id = ll.labeled_line_id JOIN attack_label al

ON al.label_id = III.label_id JOIN attack_phase ap

ON ap.phase_id = al.phase_id LEFT JOIN labeled_line_rule llr

ON llr.labeled_line_id = ll.labeled_line_id

AND llr.label_id = al.label_id WHERE ap.phase_name = 'privilege_escalation' GROUP BY
h.host_key, ae.timestamp, ae.type,
    am.op, am.acct, am.exe, am.terminal;
```

DASHBOARD PANEL 3 · SCATTER

3. Privilege escalation timeline (su then sudo)

Nine-event chain on `intranet_server` between 2022-01-24 04:37:40 and 04:38:06 UTC. Source: Alembic-managed view `v_privilege_escalation_timeline`, exposed by `sql/queries/advanced/naman_view_privilege_escalation_timeline.sql`.

Audit event type	Event timestamp (UTC)	Target account / executable
USER_AUTH	04:37:40 Jan 24, 2022	jhall, /bin/su
USER_ACCT	04:37:40	jhall, /bin/su
CRED_ACQ	04:37:40	jhall, /bin/su
USER_START	04:37:40	jhall, /bin/su
USER_CMD	04:38:06	(unset), (unset)
CRED_REFR	04:38:06	root, /usr/bin/sudo
USER_END	04:38:06	root, /usr/bin/sudo
CRED_DISP	04:38:06	root, /usr/bin/sudo

DASHBOARD PANEL 3 · 9-ROW TABLE

Burst 1 (`su` to `jhall`) clusters at 04:37:40 with four PAM events; burst 2 (`sudo cat /etc/shadow` as `root`) clusters at 04:38:06 with five events. The 25.6 second gap is visible on the x-axis.

Host	Event timestamp	Audit type	PAM op	Target account	Executable	Terminal	Labels	Rules
intranet_server	2022-01-24 04:37:40 UTC	USER_AUTH	PAM:authentication	jhall	/bin/su	/dev/pts/1	attacker_change_user, escalate	attacker.escalate.audit.su.login
intranet_server	2022-01-24 04:37:40 UTC	USER_ACCT	PAM:accounting	jhall	/bin/su	/dev/pts/1	attacker_change_user, escalate	attacker.escalate.audit.su.login
intranet_server	2022-01-24 04:37:40 UTC	CRED_ACQ	PAM:setcred	jhall	/bin/su	/dev/pts/1	attacker_change_user, escalate	attacker.escalate.audit.su.login
intranet_server	2022-01-24 04:37:40 UTC	USER_START	PAM:session_open	jhall	/bin/su	/dev/pts/1	attacker_change_user, escalate	attacker.escalate.audit.su.login
intranet_server	2022-01-24 04:38:06 UTC	USER_CMD	None	None	None	pts/1	escalate, escalated_command, escalated_sudo_command	attacker.escalate.audit.sudo.command.events, attacker.escalate.audit.sudo.command.start
intranet_server	2022-01-24 04:38:06 UTC	CRED_REFR	PAM:setcred	root	/usr/bin/sudo	/dev/pts/1	escalate, escalated_command, escalated_sudo_command	attacker.escalate.audit.sudo.command.events
intranet_server	2022-01-24 04:38:06 UTC	USER_START	PAM:session_open	root	/usr/bin/sudo	/dev/pts/1	escalate, escalated_command, escalated_sudo_command	attacker.escalate.audit.sudo.command.events
intranet_server	2022-01-24 04:38:06 UTC	USER_END	PAM:session_close	root	/usr/bin/sudo	/dev/pts/1	escalate, escalated_command, escalated_sudo_command	attacker.escalate.audit.sudo.command.events
intranet_server	2022-01-24 04:38:06 UTC	CRED_DISP	PAM:setcred	root	/usr/bin/sudo	/dev/pts/1	escalate, escalated_command, escalated_sudo_command	attacker.escalate.audit.sudo.command.events

View returned 9 rows: burst 1 = `su` from `www-data` to `jhall`, burst 2 = `sudo cat /etc/shadow` as `jhall`.

Burst 1: `su` from `www-data` to `jhall` at 04:37:40 (4 PAM events). Burst 2: `sudo cat /etc/shadow` as `root` at 04:38:06 (5 events). 25.6 second gap. The dashboard reads from this view, not from a giant ad-hoc join.

Which audit types fire on intranet_server lines that carry a privilege_escalation label?

```
SELECT DISTINCT ae.type AS audit_type FROM audit_event ae JOIN host h ON h.host_id = ae.host_id WHERE
h.host_key = 'intranet_server' AND EXISTS (

SELECT 1 FROM labeled_line ll

JOIN labeled_line_label ll_label ON ll_label.labeled_line_id = ll.labeled_line_id

JOIN attack_label al ON al.label_id = ll_label.label_id

JOIN attack_phase ap ON ap.phase_id = al.phase_id

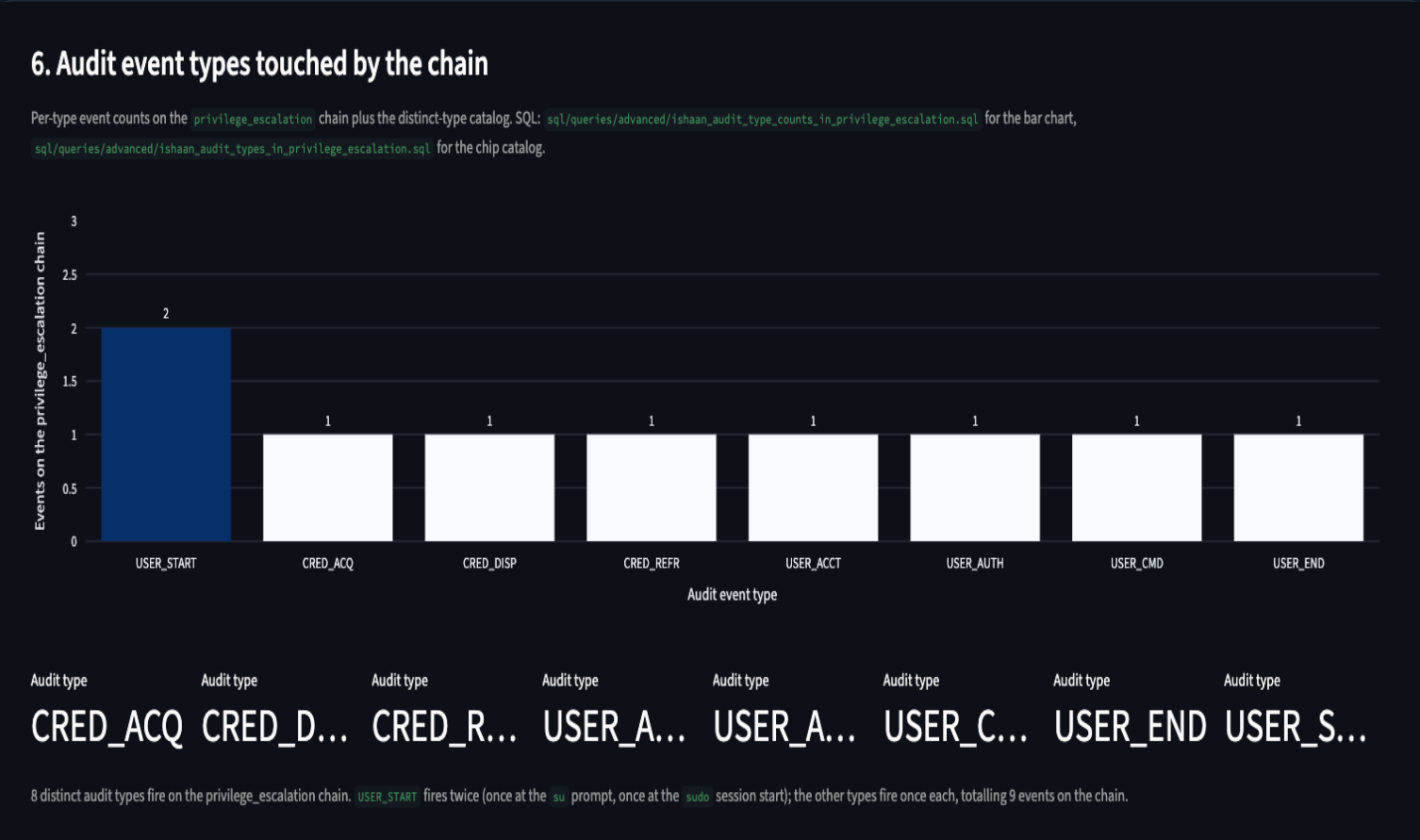
WHERE ll.source_host = h.host_key

AND ll.source_log = 'audit.log' AND ll.line_number = ae.line_number

AND ap.phase_name = 'privilege_escalation' ) ORDER BY audit_type;
```

PATTERN	DISTINCT TYPES	TOTAL EVENTS
EXISTS · 4-tbl body	8	9

DASHBOARD PANEL 6 · AUDIT EVENT TYPES TOUCHED BY THE CHAIN



8 distinct audit types fire on the chain. **USER_START fires twice** (once at the `su` prompt, once at the `sudo` session start); the other seven fire once each, totalling 9 events. A detector subscribed to those 8 types catches the chain without the full auditd firehose.

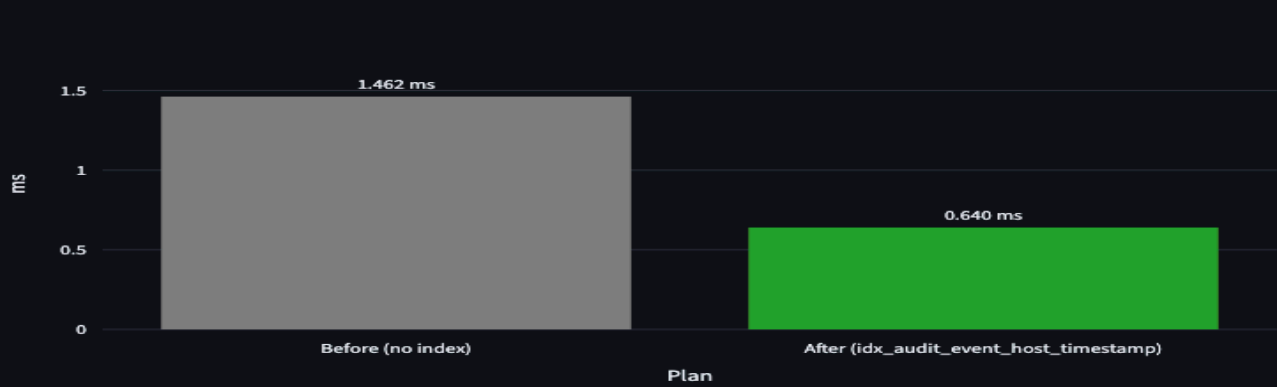
Sequential scan to index scan on the incident-response zoom.

Workload: h.host_key = 'intranet_server', ae.timestamp in 2022-01-24 04:37:30 to 04:38:15 UTC. Index: idx_audit_event_host_timestamp ON audit_event (host_id, timestamp), migration 742e860d116f. Composite index column order matches query shape: equality on host_id first, range on ae.timestamp second.

DASHBOARD PANEL 8 · INDEX IMPROVEMENT ON THE INCIDENT-RESPONSE QUERY

8. Index improvement on the incident-response query

Static EXPLAIN ANALYZE evidence captured 2026-05-01 on the russellmitchell slice (3,048 audit_event rows). SQL: sql/queries/advanced/naman_explain_index_improvement.sql . Index is Alembic managed (revision 742e860d116f, see sql/3nf/_indexes.sql).



- **Query:** incident-response zoom on `intranet_server` for the 04:37:30 to 04:38:15 UTC window on 2022-01-24.
- **Index added:** `idx_audit_event_host_timestamp ON audit_event (host_id, timestamp)` .
- **Speedup:** 1.462 ms to 0.640 ms (about 2.3 times faster).
- **Plan node:** Seq Scan (3,039 rows discarded) becomes Index Scan (Index Cond covers both equality on host_id and BETWEEN on timestamp).
- **Buffers:** shared hit drops from 141 to 33 (about 4 times less).

Plan	Execution time (ms)	Plan node	Buffers shared hit
Before (no index)	1.462	Seq Scan on audit_event	141
After (idx_audit_event_host_timestamp)	0.64	Index Scan using composite index	33

▼ Show the workload query (incident response zoom)

Exact `SELECT` measured by the `EXPLAIN ANALYZE` evidence above. Source: `sql/queries/advanced/naman_explain_index_improvement.sql` .

```
SELECT ae.timestamp,
       ae.type,
       am.op AS pam_operation,
       am.acct AS target_account,
       am.exe AS executable,
       am.terminal
FROM audit_event ae
JOIN host h ON h.host_id = ae.host_id
LEFT JOIN audit_message am ON am.event_id = ae.event_id
WHERE h.host_key = 'intranet_server'
      AND ae.timestamp BETWEEN '2022-01-24 04:37:30+00'::timestampz
                        AND '2022-01-24 04:38:15+00'::timestampz
ORDER BY ae.timestamp;
```

Streamlit + SQLAlchemy + Plotly, hitting PostgreSQL directly.

Sidebar filters: host (default intranet_server), attack phase (default initial_access + privilege_escalation), source log (multi-select).

• DEFAULT FILTERS · HOST = INTRANET_SERVER

7,748
LABELED LINES

6
DISTINCT LABELS

4
SOURCE LOGS

9
AUDIT EVENTS

• HOST FILTER CLEARED · ALL HOSTS

8,806
LABELED LINES

7
DISTINCT LABELS

7
SOURCE LOGS

9
AUDIT EVENTS

Filters

These controls drive the KPI cards at the top of the page. The seven story panels and the performance summary below the cards show the canonical narrative and stay fixed.

Host: intranet_server

Attack phase: initial_access, privilege_escalation

Source log (optional): Choose options

Story canon defaults: Host = intranet_server, Phases = initial_access, privilege_escalation, Source logs = (all)

2. Foothold accumulation across access.log.2

Cumulative count of foothold labels in 1,000-line buckets of intranet_server/access.log.2, climbing to ~7,693 by the final bucket. SQL: sql/queries/advanced/river_2/foothold_cumulative.sql.

3. Privilege escalation timeline (su then sudo)

Nine-event chain on intranet_server between 2022-01-24 04:37:40 and 04:38:06 UTC. Source: Alembic-managed view v_privilege_escalation_timeline, exposed by sql/queries/advanced/main_vtm_privilege_escalation_timeline.sql.

Host	Event timestamp	Audit type	PAM op	Target account	Executable	Terminal	Labels	Rules
intranet_server	2022-01-24 04:37:40 UTC	USER_AUTH	PAM.authentication	jhall	/bin/su	/dev/pts/1	attacker_change_user, escalate	attacker.escalate.audit.su.login
intranet_server	2022-01-24 04:37:40 UTC	USER_ACCT	PAM.accounting	jhall	/bin/su	/dev/pts/1	attacker_change_user, escalate	attacker.escalate.audit.su.login
intranet_server	2022-01-24 04:37:40 UTC	CRED_ACQ	PAM.setcred	jhall	/bin/su	/dev/pts/1	attacker_change_user, escalate	attacker.escalate.audit.su.login
intranet_server	2022-01-24 04:37:40 UTC	USER_START	PAM.session_open	jhall	/bin/su	/dev/pts/1	attacker_change_user, escalate	attacker.escalate.audit.su.login
intranet_server	2022-01-24 04:38:06 UTC	USER_CMD	None	pta1	None	None	escalate, escalated_command, escalated_sudo_command	attacker.escalate.audit.sudo.command.events, att
intranet_server	2022-01-24 04:38:06 UTC	CRED_REFR	PAM.setcred	root	/usr/bin/sudo	/dev/pts/1	escalate, escalated_command, escalated_sudo_command	attacker.escalate.audit.sudo.command.events
intranet_server	2022-01-24 04:38:06 UTC	USER_START	PAM.session_open	root	/usr/bin/sudo	/dev/pts/1	escalate, escalated_command, escalated_sudo_command	attacker.escalate.audit.sudo.command.events
intranet_server	2022-01-24 04:38:06 UTC	USER_END	PAM.session_close	root	/usr/bin/sudo	/dev/pts/1	escalate, escalated_command, escalated_sudo_command	attacker.escalate.audit.sudo.command.events
intranet_server	2022-01-24 04:38:06 UTC	CRED_DISP	PAM.setcred	root	/usr/bin/sudo	/dev/pts/1	escalate, escalated_command, escalated_sudo_command	attacker.escalate.audit.sudo.command.events

View returned 9 rows: burst 1 = su from www.data to jhall, burst 2 = sudo cat /etc/shadow as jhall.

PANELS 2 TO 3 · CUMULATIVE FOOTHOLD & SU-THEN-SUDO TIMELINE

Filters

These controls drive the KPI cards at the top of the page. The seven story panels and the performance summary below the cards show the canonical narrative and stay fixed.

Host: intranet_server

Attack phase: initial_access, privilege_escalation

Source log (optional): Choose options

Story canon defaults: Host = intranet_server, Phases = initial_access, privilege_escalation, Source logs = (all)

4. Where else does the attack leave traces

Horizontal bar chart showing the distribution of audit events across different hosts and categories. The x-axis represents the count of events, and the y-axis lists various audit categories.

5. Detection rules ranked by lines triggered

Horizontal bar chart showing the top detection rules based on the number of lines they triggered. The x-axis represents the count of lines triggered, and the y-axis lists the detection rules.

PANELS 4 TO 5 · CROSS-HOST TRACES & RULE RANK

Filters

These controls drive the KPI cards at the top of the page. The seven story panels and the performance summary below the cards show the canonical narrative and stay fixed.

Host: intranet_server

Attack phase: initial_access, privilege_escalation

Source log (optional): Choose options

Story canon defaults: Host = intranet_server, Phases = initial_access, privilege_escalation, Source logs = (all)

Privilege Escalation on intranet_server

Filtered key counts: 8,806 labeled lines, 7 distinct labels, 7 source logs, 9 audit events.

3. Foothold-to-escalate journey on access.log.2

Horizontal bar chart showing the distribution of audit events across different hosts and categories. The x-axis represents the count of events, and the y-axis lists various audit categories.

2. Foothold accumulation across access.log.2

Cumulative count of foothold labels in 1,000-line buckets of intranet_server/access.log.2, climbing to ~7,693 by the final bucket. SQL: sql/queries/advanced/river_2/foothold_cumulative.sql.

FILTER DEMO · HOST CLEARED · KPIS UPDATE LIVE

Filters

These controls drive the KPI cards at the top of the page. The seven story panels and the performance summary below the cards show the canonical narrative and stay fixed.

Host: intranet_server

Attack phase: initial_access, privilege_escalation

Source log (optional): Choose options

Story canon defaults: Host = intranet_server, Phases = initial_access, privilege_escalation, Source logs = (all)

6. Audit event types touched by the chain

Horizontal bar chart showing the distribution of audit events across different hosts and categories. The x-axis represents the count of events, and the y-axis lists various audit categories.

7. Busiest audit day on intranet_server

Horizontal bar chart showing the distribution of audit events across different hosts and categories. The x-axis represents the count of events, and the y-axis lists various audit categories.

8. Index improvement on the incident-response query

Horizontal bar chart showing the distribution of audit events across different hosts and categories. The x-axis represents the count of events, and the y-axis lists various audit categories.

PANEL 7 · BUSIEST AUDIT DAY · ATTACK DAY = 3/4

Filters

These controls drive the KPI cards at the top of the page. The seven story panels and the performance summary below the cards show the canonical narrative and stay fixed.

Host: intranet_server

Attack phase: initial_access, privilege_escalation

Source log (optional): Choose options

Story canon defaults: Host = intranet_server, Phases = initial_access, privilege_escalation, Source logs = (all)

8. Index improvement on the incident-response query

Horizontal bar chart showing the distribution of audit events across different hosts and categories. The x-axis represents the count of events, and the y-axis lists various audit categories.

PANEL 8 · INDEX IMPROVEMENT · EXPLAIN SUMMARY

Attack day on intranet_server ranks 3 of 4 days by raw audit volume. *Volume-only detection misses it; the dashboard surfaces the chain anyway.*

10 / 11 · ISHAAN · 1:30

INTERACTIVE DASHBOARD

Three findings the schema lets us defend.

01

Volume is not coverage.

Privilege escalation has only 9 audit events, yet it is the most consequential phase of the attack.

02

Volume-only detection misses the attack.

Attack day ranks 3 of 4 by raw audit volume on intranet_server. The chain hides inside baseline noise.

03

Relational design enables cross-source reconstruction.

Host inventory + audit events + labels + rules in one query is what made the 9-event chain readable.

DELIVERABLES

GITHUB REPO

22-table 3NF schema · Alembic migrations · 23 SQL queries · Streamlit dashboard · README run steps.

TECHNICAL REPORT

10-section PDF; full SQL portfolio in section 6.

DASHBOARD DEMO

Live Streamlit app with screen-recording fallback.

WHAT HAPPENED NEXT

Exfiltration on internal_share is documented in the report appendix.

FLOOR OPEN

Questions?

Naman Sudan River
Roseveare-Hunt
Ishaan Shetty